Video Transcript

Video 9: Applying Reinforcement Learning: The Frozen Lake Game (01:24)

Now that you've done some exercises in reinforcement learning and you understand how the Q-Matrix works, take a look at this quick example here called Frozen Lake. It's a fun game that you can find on the web. In this game, the person needs to navigate across the lake without falling in these holes. So, part of the lake is frozen and part of the lake has melted, and we need to reach this goal.

You'll be familiar with the Q-learning matrix. So, in this case, there are 16 different states and each state has four actions. And so, we've got a 16 by 4 Q-learning table. There's some randomness injected into the game, so you won't always win. So, run the game, change some of the hyper parameters such as the gamma and the alpha and see how good your robot is. Here, I reached the goal a few times, but I fell into the hole just as many. Okay. Obviously, I need to have it learn better. So, enjoy Q-learning with the Frozen Lake. Bye for now.

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